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Research on the connectivity between injection and production wells in the Hongqian in situ combustion project in Xinjiang based on machine learning

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ABSTRACT

Accurate assessment of injection-production connectivity is critical for optimizing the in-situ combustion process. Conventional methods often focus on individual geological or dynamic parameters, lacking a comprehensive evaluation framework. This study, centered on the Hongqian-1 Well Block pilot area, processed all logging curves through noise reduction and segmentation, then quantified structural similarity between injection and production well curves as a static similarity coefficient to reflect geological connectivity. Machine learning algorithms were used to identify the most influential dynamic features. These features were integrated with the static coefficient to establish a hybrid geological-dynamic model that generated connectivity coefficients Validation via production trends and tracer tests confirmed the model's accuracy, demonstrating that the calculated inter-well connectivity aligns with actual production behavior and tracer results. The static coefficient effectively characterizes reservoir heterogeneity and significantly influences connectivity patterns. This method provides a time-efficient, reliable basis for understanding in-situ combustion connectivity and guiding injection-production adjustments.

1. Introduction

In-situ combustion (fire flooding) is one of the effective methods for the development of heavy oil reservoirs. This process involves injecting large volumes of air into formations, where oxygen acts as an oxidizer to sustain underground combustion reactions (Yuan et al., 2019). Compared to water flooding, it exhibits greater complexity due to interacting factors including gas injectivity, combustion dynamics, and coupled physicochemical reactions (Wang et al., 2019a). Assessing injection-production connectivity and tracking combustion front propagation constitute critical priorities for optimizing in-situ combustion operations.

Traditional methods for diagnosing inter-well connectivity include numerical simulation (Vargas-Guzman et al., 2009), tracer testing (Yousef et al., 2009), and interlayer comparison (Wang et al., 2017). However, these methods all have problems such as being time-consuming or costly to varying degrees. He et al. (2023) established a model capable of predicting production through in-situ combustion experiments by using air injection volume data and production data. However, their model requires stable production data and does not account for connectivity between injection and production wells. Shibao Y et al. (Yuan et al., 2018) further analyzed the production status at the in-situ combustion site through core experiments on site, which is helpful for further evaluating the in-situ combustion production process. However, this approach is economically costly and cannot characterize connectivity during the in-situ combustion production stage. Peng et al. (2022) studied the injection-production dynamics of in-situ combustion by using numerical simulation and simulated the production situations at different stages of in-situ combustion. Although the numerical simulation method is relatively efficient, it still requires a certain learning cost in terms of technology, and its operation is complex, making it difficult to accurately quantify the correlation status between injection and production wells in in-situ combustion.

Recent advancements in applying machine learning (Nayyef et al., 2024a) (Tahri et al., 2016a) to thermal recovery techniques in petroleum engineering have demonstrated significant potential. Muhammad E.D.R. et al. (Nayyef et al., 2024b) employed a random forest algorithm to perform feature importance ranking of reservoir variables, thereby optimizing steam injection prioritization in enhanced oil recovery (EOR) processes. This data-driven approach offers a systematic framework for

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reservoir management. Furthermore, Guoying W. et al. (Tahri et al., 2016b) established a robust linear regression model correlating thermal conductivity with acoustic velocity in oil shale under thermal recovery conditions. Their findings provide critical insights for thermo-mechanical modeling in heavy oil reservoirs. The rapid and cost-effective calculation of inter-well connectivity using production data has emerged as a research hotspot in recent years. Yu et al. (Li et al., 2023a) employed a random forest algorithm to model heavy oil steam stimulation processes, utilizing Pearson correlation coefficients for feature selection to predict steam channeling. This approach accelerated the identification of inter-well steam breakthrough channels compared to conventional modeling methods while incorporating geological features (e.g., permeability) into feature analysis. However, the reliance on Pearson correlation analysis may overlook essential features due to its inability to capture nonlinear relationships between variables.However, most related studies focus on water flooding development. When analyzing inter-well connectivity, there is generally a situation where static geological data or dynamic production data are solely relied upon, and the advantages of integrating and applying both have not been fully exploited. Rogulina et al. (2022) provided a basis for predicting well logging data and optimizing logging measurement selection. They trained using the logistic regression and XGBoost algorithms, with the characteristics of well interval logging data as input, and output the similarity of lithology among multiple well intervals,the feature quality metric reached 0.84 (ROC AUC). However, it overly relies on the training model, has poor interpretability, and it is difficult to understand the decision-making process of the model. Joshi et al. (2023) employed linear regression to predict acoustic logging data and subsequently applied K-means clustering to identify lithological types including siltstone, coal, and sandstone. The model achieved training set $R^2$is 0.95 and test set $R^2$is 0.96, demonstrating robust predictive performance across both datasets. However, the analysis is only based on single logging data, failing to fully explore the potential features of the data and having insufficient adaptability to complex geological conditions. Gong et al. (2021) took the Su X block in the eastern Sulige Gas Field as an example, combined logging and production data, established a model of the interference degree of oil wells, and determined the continuity and connectivity probability by making a well-tie profile to observe the sand body distribution. However, the characteristic parameters it considered were numerous and complex, with strong limitations and interferences, and it failed to clearly identify the main features affecting the connectivity between interfering wells.Using dynamic production data, some scholars analyzed and characterized the inter-well connectivity between injection wells and production wells based on machine learning regression algorithms (Albertoni et al., 2003) (Liang et al., 2007), Pearson correlation analysis method (Refunjol, 1996), etc. Factors such as the stability of the water injection process and the opening and closing of wells will affect the calculation accuracy. Heffer et al. (1997) established a connectivity model between one injection well and one production well using the Spearman correlation coefficient method and studied the reservoir connectivity by using the dynamic data fluctuations of injection and production wells. However, it did not consider the time lag of the water injection signal (Yousef et al., 2006). In view of the physical property changes during the oilfield development process and combined with the advantages of machine learning optimization calculation results, Tiab et al. (Dinh and Tiab, 2008) established a connectivity inversion model using the bottom hole flowing pressure of injection and production wells. However, the model requires injection and production wells to maintain stable production, which limits its application. Li Xianwen (Li et al., 2013), Chen Cunliang (Chen, 2020) developed a novel interwell connectivity model integrating genetic algorithm-based parameter inversion and tracer response evaluation using injection-production data. Their approach demonstrated field efficacy through foam profile control treatments, achieving 6.8% water-cut reduction and $1.53 \times 10^4 m^3$incremental oil recovery. However, only the response relationship between water injection and liquid production was considered in the calculation process, without considering the impact of other factors on the calculation results.All the above domestic and foreign studies take water injection development as the research object, and mostly take single production data or geological data as the research object. The applicability of combining static and dynamic data to the injection-production process of in-situ combustion needs to be further tested (Hittawe et al., 2024).

Addressing limitations in interwell connectivity analysis—including reliance on single dynamic (e.g., production rates, pressure) or static (e.g., permeability, porosity) factors, unclear key correlations, and high computational complexity—this study leverages dynamic production data from the steady fire flooding stage in the Hongqian-1 pilot zone, Xinjiang. In contrast to existing machine learning models that often focus on a single data type, our work establishes a novel hybrid approach. Methodological advances include: (a) Integrated analysis of static logging data (for injector-producer similarity) and dynamic engineering data (production history, operational parameters) as model inputs, creating a unified geological-dynamic (G&D) framework; (b) Machine learning to identify key production indicators driving injector-producer interactions via regression quantification; (c) Quantitative evaluation of connectivity dynamics by linking geological traits (e.g., heterogeneity) to operational parameters (e.g., injection pressure gradients). This model is specifically designed for the in-situ combustion process, addressing challenges such as gas compressibility and nonlinear combustion fronts. Field validation confirms the framework's accuracy, offering actionable insights to optimize fire flooding via targeted well interventions and injection adjustments.

2. Overview of the research area and data processing

2.1. Overview of the research area

The pilot research area of Well Block Hongqian 1 is located at the eastern section of the northwestern margin of the Junggar Basin. Its structural form is a monocline gently dipping to the southeast, with a formation dip angle of 4°. The target layer of the research area is the Badaowan Formation. The average burial depth of the oil reservoir is 540 m, and the average effective thickness is 9.6 m. Overall, it is mainly characterized by channel microfacies deposits, and locally by channel bar microfacies deposits (Muhetaer et al., 2021). The reservoir as a whole belongs to a medium-porosity and medium-permeability reservoir with a normal temperature and pressure system. It had previously undergone the development processes of steam huff and puff and steam flooding. In 2009, in-situ combustion development tests were carried out on Well Groups 008h and 010h in the research area. In general, in the research area, the remaining oil saturation is relatively high near the southwest, while it is relatively low in the middle and eastern parts. The construction situation is complex. There are currently three gas injection wells and three rows of production wells (Fig. 1). After the in-situ combustion development in this research area, certain achievements have been made, and the production data is relatively complete.

During the monitoring process of the in-situ combustion test, it was found that some wells had problems such as low liquid production and water cut, low tracer concentration, and well shut-in during the test period. As a result, the results could not fully reflect the inter-well connectivity characteristics during the test stage. Through the analysis of the facies belt and core in the pilot area, Well Group 008h is located within the mid-channel bar microfacies, with weak heterogeneity. In contrast, Well Group 010h is located within the braided river facies microfacies, with strong heterogeneity and a permeability ratio greater than 20. In the subsequent in - situ combustion production process, it was found that the fire front advanced uniformly at a moderate speed in Well Group 008h, and the over - covering was not severe. However, in Well Group 010h, there were significant differences in the fire front advancement, and gas channeling was severe. Overall, the Well Group 008h in the southwest consists of underwater distributary channels sand

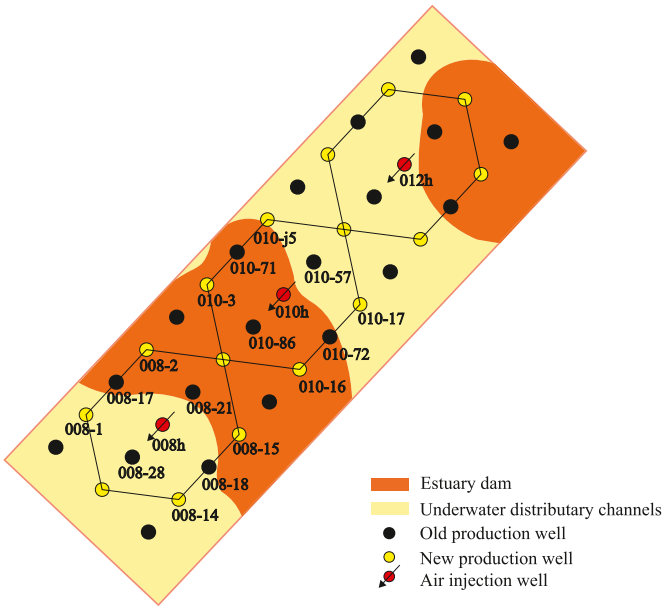


Fig. 1. Well location distribution map of the research area.

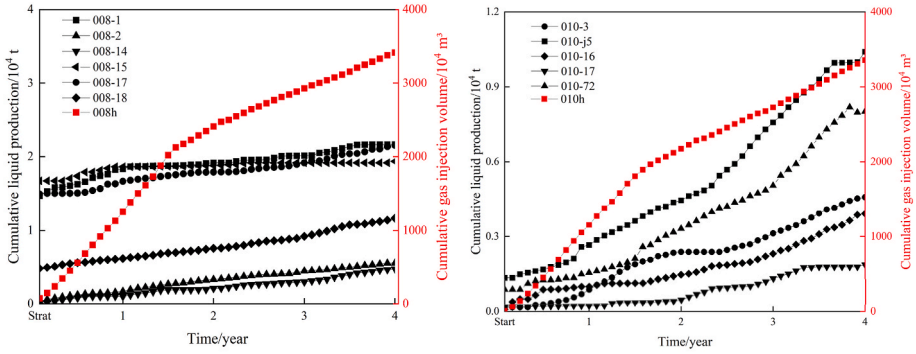
bodies, with good overall physical properties, an average porosity of 23.1%, an average permeability of 650 mD, and moderate heterogeneity. The Well Group 010h in the middle consists of estuary dam sand bodies. Well 010-72 has the best physical properties, with a permeability of over 1500 mD and the strongest heterogeneity.The injection situation of injection wells and the liquid production situation of production wells in the four - year stable production stage of in - situ combustion in the two well groups are shown in Fig. 2. As can be seen from Fig. 2, affected by the previous development of the block and formation heterogeneity, etc., in the same block, for production wells in the same well group and those in different well groups, when the gas injection volume has little difference, the liquid production volume shows obvious differences. Most production wells in Well Group 008h are in a relatively stable state, while the liquid production volume of each production well in Well Group 010h fluctuates significantly. This is caused by the differences in the correlation between injection wells and production wells under the condition of formation heterogeneity.

The conventional methods for calculating inter-well connectivity through injection-production correlation include the grey correlation analysis method of injection-production data (Dinh and Tiab, 2008) and the multiple linear regression method. The production data of the stable production stage of in-situ combustion in the 008h and 010h well groups in the pilot research area of Well Block Hongqian 1 within four years were selected. The production data of 1 injection well and 5 production wells in the 008h well group, as well as the production data of 1 injection well and 4 production wells in the 010h well group were chosen. Then, the grey correlation analysis method and multiple linear regression were used to analyze the injection-production correlation of each well group.

The grey correlation analysis method was used to conduct a correlation analysis on the injection-production well groups in Well Group 010h. The grey correlation degree between the liquid production volume of the production wells in the well group and the injection volume of the gas injection well is shown in Fig. 3(a). The magnitude of the grey correlation degree indicates the correlation degree of the injection-production data. It can be seen that the injection-production correlation degrees of the four production wells do not differ much, which is inconsistent with the actual production situation. This indicates that the injection-production relationship in in-situ combustion is not a simple linear correlation situation that can be studied by grey correlation analysis. In 2003, Albertoni and Lake (Albertoni et al., 2003) proposed that the multiple linear regression (MLR) model could be used to quantify and characterize inter-well connectivity. They regarded the injection well, the production well and the inter-well channel as a system. The output response of a production well corresponds to the input incentives of all surrounding injection wells, and the regression weight between the production well and the injection well is the inter-well connectivity. The datasets of liquid production and gas injection indicators were selected to establish a multiple regression model, and the inter-well connectivity coefficients of the injection-production wells in Well Groups 008h and 010h were obtained through regression calculation using the least squares method (Liu, 2009), as shown in Fig. 3(b).

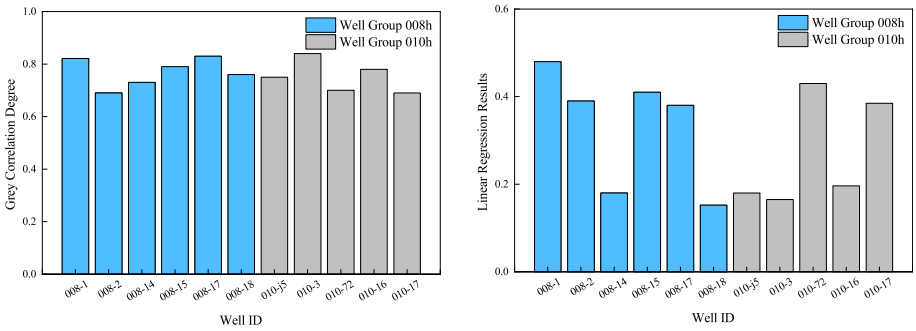
Compared with the actual field analysis and the actual liquid production data (Fig. 2), it is found that it is difficult to distinguish the inter-well connectivity status through the grey correlation results (Fig. 3(a)). Moreover, the calculation results of Well 010-j5 and Well 008-2 are on the low side, while those of Well 008-17, Well 010-16 and Well 010-3 are on the high side, which is inconsistent with the actual production situation. There is a great contradiction between the linear regression results (Fig. 3(b)) and the actual production situation. For wells with better production, the quantification results of their connectivity are poor, which is inconsistent with the actual production situation, such as Well 008-18, Well 010-j5, Well 010-72, Well 010-17 and Well 010-3. It is analyzed that in-situ combustion is different from water flooding. The compressibility of gas leads to the insignificant impact of in-situ combustion on the liquid production volume after gas injection. Therefore, it is impossible to simply perform regression fitting from the liquid production volume and the injection volume to obtain relatively accurate results. It is necessary to consider using other relevant characteristic data to participate in the fitting.

Consequently, in this paper, geological data and engineering data are comprehensively selected, combining static geological data and



(a) 008h well group (b) 010h well group

Fig. 2. Injection and production data of research area well groups.



(a) Results of Grey Correlation Degree (b) Results of Linear Regression

Fig. 3. Results of injection-production well correlation analysis calculated by conventional methods.

dynamic production data, integrating various features to calculate the connectivity between injection and production wells during the in-situ combustion production process.

2.2. Logging curve data processing

Different logging curves can obtain different reservoir geological information in the near-wellbore area at a specified depth, including rock density, porosity and oil-bearing physical properties of the reservoir, etc. When adjacent wells show similar reservoir physical properties, it often means that the formations they are located in have common geological features, and the patterns of the logging curves are also more similar. By comparing the structural characteristics of the logging curves and calculating the degree of similarity, the static similarity coefficient between injection and production wells can be obtained.

Different types of logging curves can provide different formation information (Li et al., 2023b). To facilitate the analysis of the similarity of the formations between injection and production wells in in-situ combustion, three types of logging curves, namely gamma ray (GR), acoustic (DT), and resistivity (RT), are selected for calculation and analysis, which can respectively reflect the lithological characteristics, reservoir physical properties and reservoir oil-bearing conditions of the formations.

2.2.1. Logging curve smoothing preprocessing

There are many fluctuating noises in the original logging curves. Directly using the original data will lead to being trapped in local noise interference. Smoothing preprocessing can weaken these fluctuations and is beneficial for highlighting the overall useful information for subsequent calculations (Advancing Arabic Handwritten Digit Recognition, 2024).

Firstly, the Savitzky-Golay filter (Dombi and Dineva, 2020) (abbreviated as S-G filter) is used to conduct noise reduction processing on the logging data. Its process is shown in Fig. 4. The S-G filter can keep the shape and width of the signal unchanged while suppressing the fine features, which is convenient for use in the subsequent structural and trend comparative analysis of the logging curves.

The S-G filter mainly realizes the noise reduction process by adjusting the window size and the polynomial order (Mohammad et al., 2020). The window value is usually selected as an odd number to ensure the symmetry of the center point, and the polynomial is adjusted according to the processing effect. An overly high polynomial order and an overly small window are prone to cause overfitting, and the filtered curve will have many fine features and it will be difficult to grasp the overall trend. Through cross-validation, choosing the order of 2 or 5 and the window number of 15 or 25 can meet the requirements. In this paper, the window size of the S-G filter is selected as 15, and the polynomial order is 2. As shown in Fig. 5, the parameters used in this paper have reduced the noise in the details of the original logging data to a large extent after noise reduction and retained the significant fluctuation amplitude and trend of the logging curve.

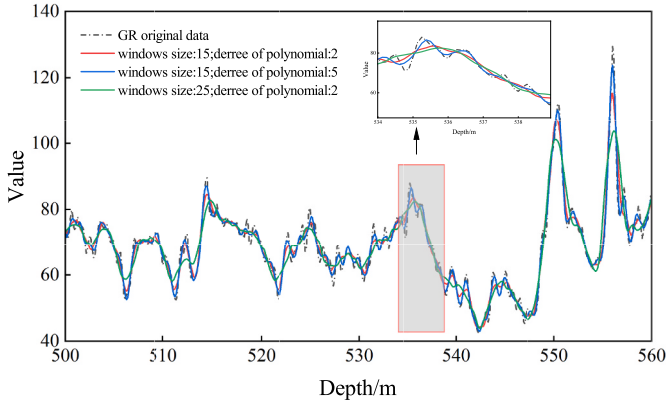


Fig. 5. Comparison of GR logging curve of well 008h before and after suppressing fine features.

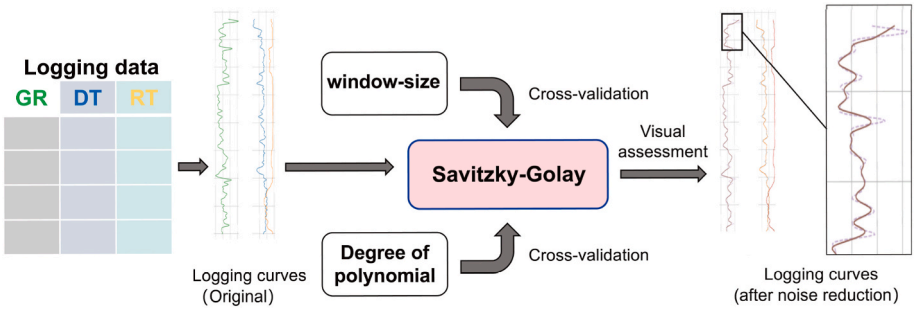


Fig. 4. Logging curve data processing flow.

2.2.2. Split of logging curves

In the actual formation correlation process, a representative curve pattern and structural pattern of a certain well are often extracted as a unit, and then similar units are searched for within a similar depth range in other wells. To restore the formation correlation process, the method of splitting logging curves is introduced. That is, the parts of the logging curves with typical features are cut out as units, and then similar units are searched for and compared within the depth range of other wells. Judging whether there are similar features in the formations within a certain depth range between two wells, that is, through the similarity degree between the logging curves, can reflect the static connectivity status between the two wells.

When splitting the logging data, appropriate splitting intervals are selected for cutting according to their significant structures. Taking the GR curve of Well 008h as an example, to ensure that each group of split data points is uniform and can contain the complete features of the curve to the greatest extent, every 36 data points are grouped for splitting. In the depth range from 500 m to 560 m, a total of 15 units are split. Each group of logging curves after splitting should retain an appropriate logging curve structure to avoid significant loss of logging data features and obtain relatively accurate similarity results.

2.3. Production dynamic data processing

There are many types of reservoir development dynamic data, such as gas injection volume, production time, oil pressure and other data. However, the data actually recorded in the oilfield often have problems such as anomalies, missing values or other issues. If machine learning is carried out based on the original data, the learning process of the model will be uncontrollable and situations like excessive errors in the results will occur. Data preprocessing is to analyze, classify, eliminate, supplement and correct the original and complicated production data, so that the original dynamic production data can become an ideal dataset for machine learning modeling. Taking Well Group 008h in the in-situ combustion pilot research area of Well Block Hongqian 1 as an example in this paper, the main steps of data preprocessing are as follows:

1) Production dynamic data from the Hongqian-1 pilot test area were selected, with dataset characteristics detailed in Table 1.
2) A systematic data quality control procedure was implemented to address data anomalies and missing values. Outliers were identified through interquartile range (IQR) analysis and validated against production records to differentiate measurement errors from physically plausible extremes. Non-physical anomalies were removed based on operational constraints.For missing data imputation, the k-nearest neighbors (KNN) algorithm was employed. This approach utilizes local similarity patterns within the dataset to estimate missing values while maintaining inherent data structures. The method demonstrated particular efficacy for time-series production data by preserving temporal correlations and minimizing imputation bias.
3) Take the monthly oil production volume as the target variable, and convert the original data into a binary classification: low production (coded as 0) and non-low production (coded as 1). The classification threshold of 30 tons/month was determined based on the historical production performance analysis of the Hongqian-1 pilot area and aligns with the operational benchmark used in field management to define underperforming wells. Select the remaining 7 datasets as the characteristic variables.

3. Construction of the G&D-Principal feature injection-production correlation model

3.1. Model principle

When a production well is affected by multiple surrounding injection wells, the injection volume of the i-th injection well under the multiple linear regression model is expressed as:

$$\widehat{q}_j(n) = \beta_{o_j} + \sum_{i=1}^{I} \beta_{ij}\, i_i(n) (j = 1, 2, \cdots, N) \tag{1}$$

In Equation (1): $i_i(n)$ represents the injection volume of the i-th injection well at time step n; β_{o_j} is the constant term, the injection-production imbalance coefficient; β_{ij} is the multiple regression weight between the j-th production well and the i-th injection well; $\widehat{q}_j(n)$ is the liquid production volume of the j-th production well at time step n.

During the in-situ combustion injection-production process, the situation of one injection well corresponding to multiple production wells often occurs, and the adjacent injection wells are relatively far apart. Drawing on the idea of establishing the multiple linear regression model (Albertoni et al., 2003), assuming that the j-th production well is only affected by the nearest i-th injection well, then Equation (1) becomes Equation (2) for this situation:

$$\widehat{q}_j(n) = \beta_{o_j} + \beta_{ij}\, i_i(n) \tag{2}$$

In a system, the output information may be generated under the combined action of multiple input information. In the case where the correlation between the gas injection volume and the liquid production volume is not obvious, the liquid production volume of a single well can be taken as the dependent variable, and the injection volume and the principal feature variables that affect the liquid production volume can be taken as the independent variables to establish the principal feature injection-production correlation model based on geological and dynamic features (abbreviated as G&D), as shown in Equation (3).

$$\widehat{q}_j(n) = \beta_{o_j} + \beta_{g(ij)}\beta_{ij} i_i(n) + \sum_{m=1}^{M} (\lambda_m A_m(n)) \tag{3}$$

In the equation, $i_i(n)$ represents the injection volume of the i-th injection well at time step n; β_{o_j} is the constant term, the injection-production imbalance coefficient; $\beta_{g(i,j)}$ is the obtained static similarity coefficient between the j-th production well and the i-th injection well; β_{ij} is the dynamic response coefficient between the j-th production well and the i-th injection well; $\beta_{ij}\beta_{g(i,j)}$ is the connectivity coefficient between the j-th production well and the i-th injection well; $\widehat{q}_j(n)$ is the liquid production volume of the j-th production well at time step n; m is a major correlation feature obtained through machine learning feature screening; M is the number of major correlation features obtained through machine learning feature screening; λ is the regression weight of each major correlation feature; A_m is the main correlation features identified by machine learning, excluding $i_i(n)$.

The principle of Equation (3) is based on the multiple linear regression principle. Among them, the injection volume, as an input, directly affects the magnitude of the liquid production volume. The other m principal feature variables can improve the fitting accuracy of the principal feature injection-production correlation model, and the number of correlation factors should not be too large (Wang et al., 2019b) and needs to be determined through screening and calculation. In order to select the principal features from numerous datasets, machine learning can be used for screening and determination.

For the research on the inter-well connectivity in in-situ combustion, the G&D-Principal Feature Injection-Production Correlation Model for inter-well connectivity between injection and production wells is

established. Machine learning is used to screen the key feature datasets that affect the injection-production connectivity relationship. Then, the screened datasets are used to calculate the injection-production correlation weights (i.e., inter-well connectivity coefficients). Finally, production feature analysis, tracer monitoring and other methods are used to verify the correctness of the results. The corrected model can be used to adjust the subsequent in-situ combustion production. The specific process is shown in Fig. 6.

3.2. Calculation of static similarity coefficient

The static similarity coefficient serves as a quantitative characterization of reservoir petrophysical characteristics in fire-flooded injector-producer wells. It is derived from similarity calculations of corresponding well logging curves between these wells. By analyzing the structural and trend similarities in the curves, the inter-well geological characteristics can be comparatively evaluated (Hittawe et al., 2015). It is obtained through similarity calculation based on the logging curves of the corresponding injection-production wells in in-situ combustion. Its result represents the similarity degree of the reservoir physical properties of the corresponding injection-production wells in in-situ combustion within a certain depth range. The larger the value is, the more similar the physical properties of the corresponding reservoirs of the injection well and the production well are, and the better the connectivity status is.

The calculation process of the static similarity coefficient is shown in Fig. 7. Within the same well group, taking the gas injection well in the well group as the reference, the processed (denoised and segmented) logging curve is used as the reference curve. The logging curves of the remaining production wells are also processed in the same way to serve as the control curves. A structural correlation analysis was conducted by computing the Structural Similarity Index (SSIM) between the production well comparison curve and the baseline curve, assessing similarity within each unit segment across the analyzed interval (Zhang et al., 2020). Following each assessment, both curves were shifted downward by one unit for subsequent analysis.

The weighting scheme applied to the well logs in Equation (4)—specifically, the assignment of 0.6 to the resistivity (RT) log and 0.4 to the acoustic (DT) log—is expert-driven and grounded in established petrophysical principles. This allocation reflects the relative diagnostic importance of each curve for characterizing the target interval. The natural gamma ray (GR) log is assigned the highest priority because of its superior ability to identify lithology (Shen et al., 2010), which forms the foundation for any further reservoir comparison. Between the two remaining curves, the RT log receives a higher weight because it exhibits greater sensitivity to reservoir fluid content and effective porosity, whereas the DT log is more influenced by rock-matrix properties. This weighting strategy follows the feature-domain weighting principle commonly used in structural similarity analysis and ensures that the resulting static similarity coefficient effectively captures key subsurface heterogeneities (Li and Bovik, 2009). Although the specific weights were not empirically tuned in this study, they are supported by domain knowledge and were indirectly validated through the model's overall performance and agreement with field observations.

$$\beta_{g(i,j)} = S_{GR} \times (0.6S_{RT} + 0.4S_{DT}) \tag{4}$$

In the equation, $\beta_{g(i,j)}$ represents the static similarity coefficient between the i-th production well and the corresponding j-th injection well, and $\beta_{g(i,j)} \in [0,5]$; S_{GR} is the structural similarity coefficient of the GR curve between the production well and the corresponding injection well; S_{RT} is the structural similarity coefficient of the RT curve between the production well and the corresponding injection well; S_{DT} is the structural similarity coefficient of the DT curve between the production well and the corresponding injection well.

Calculate the static similarity coefficients of the injection-production wells in Well Group 008h and Well Group 010h. As shown in Table 2, within the same well group, there are relatively obvious differences in the static similarity coefficients of the logging curves. The connectivity status of the injection-production wells within the same well group can be preliminarily judged. The larger the value is, the better the geological connectivity status between the injection-production wells in in-situ combustion within the well group is. However, as the in-situ combustion production progresses, the connectivity status between the injection-production wells may change. Minor differences in the static similarity coefficients cannot accurately represent the actual connectivity between the injection-production wells. Therefore, in this paper, combined with the in-situ combustion production data, further research on the inter-well connectivity of the injection-production wells in the pilot research area is carried out.

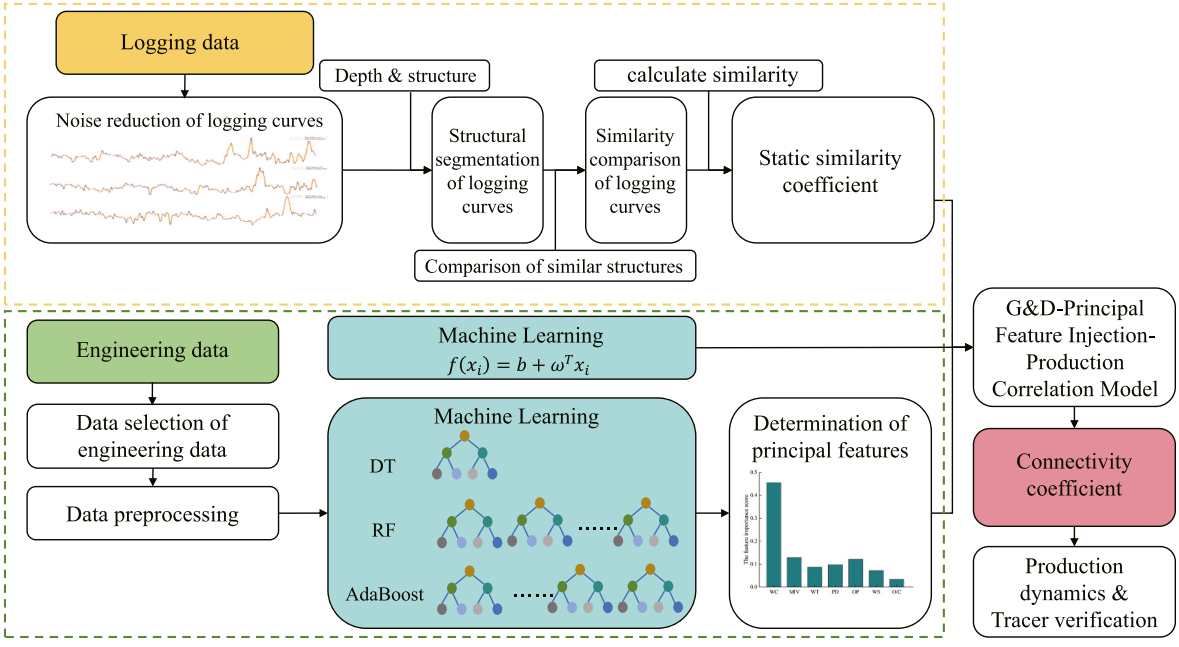


Fig. 6. Flow of research on inter-well connectivity of injection-production wells in in-situ combustion.

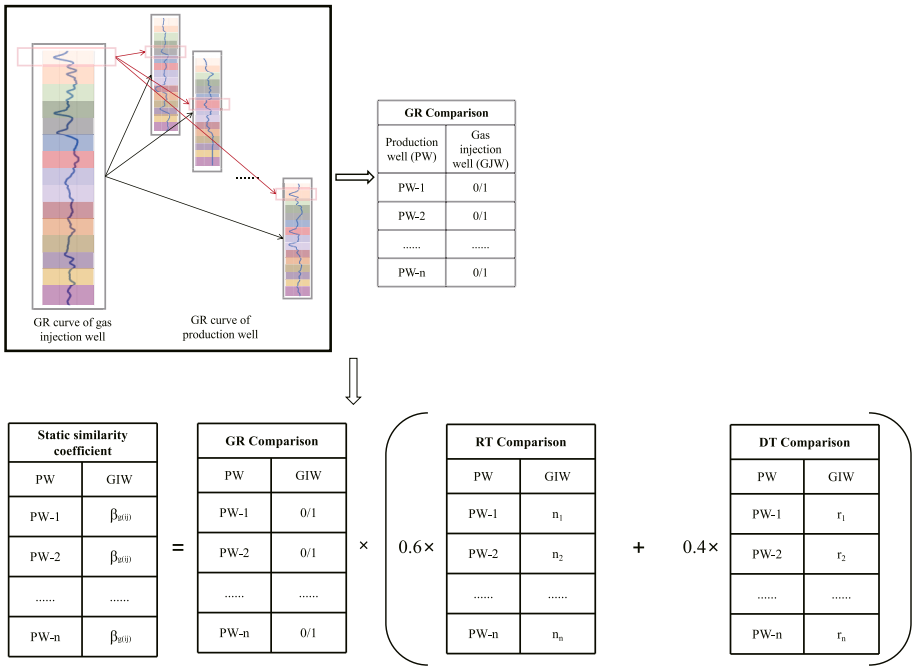


Fig. 7. Flow chart of calculating static similarity coefficient.

Table 1
Feature selection for data samples.

Injector Wells	Producer Wells	Dynamic Features	Production time (months)
008h	008-1	Monthly oil production (MOP), Water-cut (WC), Monthly Gas Injection (MIV),Wellhead Temperature (WT),Production Days (PD),Oil Pressure (OP),Well Spacing (WS),On/Off (O/C)	1~48
	008-2		
	008-14		
	008-15		
	008-17		
	008-18		
010h	010-j5		
	010-3		
	010-72		
	010-16		
	010-17		

Table 2
Calculation results of static similarity coefficients of each well.

Gas Injection Well ID	Production Well ID	Static Similarity Coefficient $\beta_{g(i,j)}$
008h	008-1	3.06
	008-2	1.54
	008-14	1.64
	008-15	2.82
	008-17	2.61
	008-18	1.41
010h	010-j5	3.67
	010-3	0.97
	010-72	1.81
	010-16	0.97
	010-17	0.80

It should be noted that the magnitude of the static similarity coefficient can only be compared within the same well group, and the comparison of the magnitude of values in different well groups has no practical significance.

3.3. Selection of principal features based on the training accuracy of the algorithm model

In the process of machine learning research, it is necessary to select several influential indicators (principal features) from numerous indicators for research, so as to avoid cumbersome analysis work and overfitting phenomena, reduce the risk of overfitting, and ensure the controllability of model operations and the accuracy of results. This process is the principal feature selection process of the model (Chauhan and Henrietta, 2023). Screening principal features can not only reduce the workload of the subsequent machine learning process but also discover some influencing factors that are easily overlooked in daily analysis. Different machine learning algorithms have their own advantages, and it is necessary to judge which algorithm to choose based on the accuracy of the training set and the test set (Hakim, 2024). The Decision Tree, Random Forest, and AdaBoost algorithms (Ali et al., 2022) (Zifeng et al., 2023) possess inherent feature importance assessment capabilities based on their respective model mechanisms (e.g., information gain in Decision Trees, mean impurity decrease or permutation importance in Random Forests, and adjustment of weights for misclassified samples in AdaBoost). These capabilities enable the automatic identification and ranking of feature contributions directly from the data. Their robustness (achieved through ensemble strategies and randomness in Random Forests, which effectively mitigate overfitting risks) and adaptability (as seen in AdaBoost's focus on difficult samples to enhance the weight of critical features) contribute to more reliable and interpretable evaluation outcomes. Consequently, these methods facilitate efficient feature selection without relying on external filtering approaches.

When utilizing Decision Trees, Random Forests, and AdaBoost algorithms to screen primary features, the dataset was partitioned into training and test sets at an 8:2 ratio. The process for selecting primary features from the preprocessed dataset is outlined as follows:

1) Utilizing the decision tree algorithm to screen primary features involves calculating the feature importance scores of each feature during the computation process of the decision tree model. Using monthly oil production as the target variable, the initial computation with default parameters resulted in a training set accuracy of 1.0 and

a test set accuracy of 0.6, indicating significant overfitting. To mitigate severe overfitting and improve test set accuracy, pruning operations were performed on the decision tree. A grid parameter search was employed to identify the optimal maximum tree depth and maximum number of leaf nodes.The evaluation of the pruned decision tree model demonstrated that on the training data, it attained an accuracy of 0.92 and a Mean Squared Error (MSE) of 0.102. When applied to the test set, the model achieved an accuracy of 0.79 and an MSE of 0.102.

2) The dominant features were selected using the Random Forest algorithm. A 5-fold cross-validation was employed to optimize the hyperparameters of the model, which uses decision trees as base estimators. The optimal configuration was determined to be 600 estimators, a maximum depth of 10, and a maximum number of features set to the square root of the total features. The model demonstrated a mean cross-validation accuracy of 0.89 ± 0.04, indicating minimal performance fluctuation and reasonable stability. The final model achieved a training accuracy of 1.00, a test accuracy of 0.89, an Out-of-Bag (OOB) score of 0.90, and an Area Under the Curve (AUC) value of 0.89.
3) The AdaBoost algorithm was also employed to screen for primary features, utilizing a decision tree as the base estimator. Through cross-validation, the number of estimators was set to 600, with a learning rate of 0.5 and a maximum tree depth of 8. The classification performance of the final learner was evaluated using a confusion matrix and ROC curve, resulting in an AUC value of 0.869. The model achieved a training accuracy of 1.0 and a test accuracy of 0.813.

As shown in Fig. 8, the feature importance results calculated by different models vary, and the importance scores of different features also differ. For example, the Production Days (PD) item scored higher in the Decision Tree and AdaBoost models but received a lower score in the Random Forest model. By analyzing the raw data, we found that the Production Days data primarily consisted of only two types: either 30 days (or close to 30 days) or 0 days. The other two models significantly amplified the importance of the Production Days feature. Due to its inherent randomness and ensemble learning characteristics, the Random Forest tends to provide a more balanced and robust assessment of feature importance. The mechanism of Random Forest avoids over-reliance on any single feature and uncovers alternative features or more stable feature combinations.

As shown in Fig. 9, a comparison of the three algorithms on both the training and test sets reveals that the Random Forest algorithm performed best on both datasets, with accuracy rates exceeding 0.9,

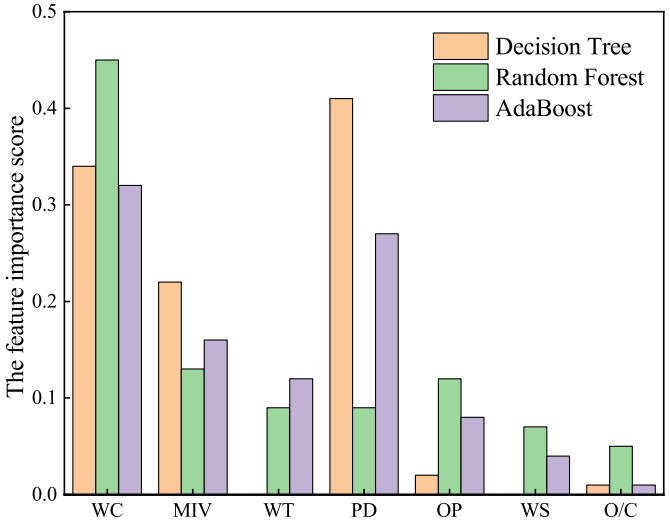


Fig. 8. Importance score of each feature of three algorithms.

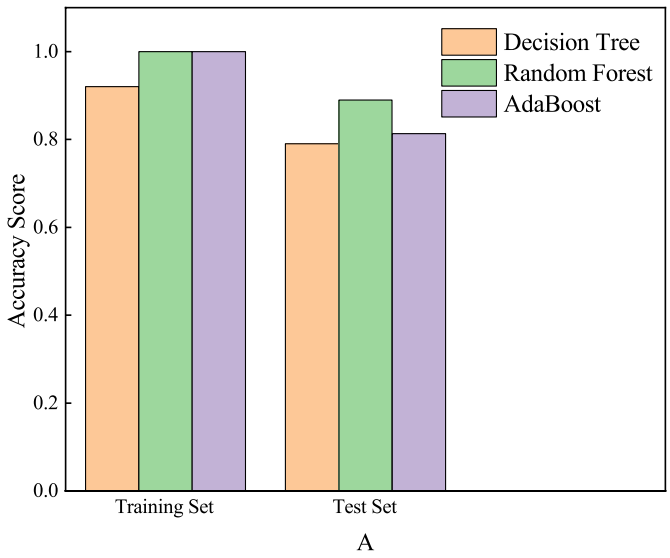


Fig. 9. Accuracy scores of three algorithms.

significantly higher than the other two algorithms. Although the algorithm achieved an extremely high score on the training set, indicating a potential risk of overfitting, its test set accuracy remained within an acceptable range, demonstrating that the model still possesses good generalization capability. Compared to other domains, Random Forest can more accurately screen key features from a large number of engineering parameters and historical data in oil well production prediction, effectively capture nonlinear relationships and interaction effects between variables, and is particularly adept at handling high-dimensional data while mitigating overfitting issues.

Based on the feature importance ranking (from high to low) provided by the random forest model, features were sequentially incorporated in varying numbers to construct the model, and the optimal size of the feature subset was determined by evaluating the model's accuracy. As shown in Fig. 10, the model's accuracy requirements could be met when the number of selected features was three. Consequently, the top three features by importance score from the random forest algorithm—water cut (WC), monthly injection volume (MIV), and oil pressure (OP)—as shown in Fig. 8, were selected as the primary features influencing oil production in the dataset.

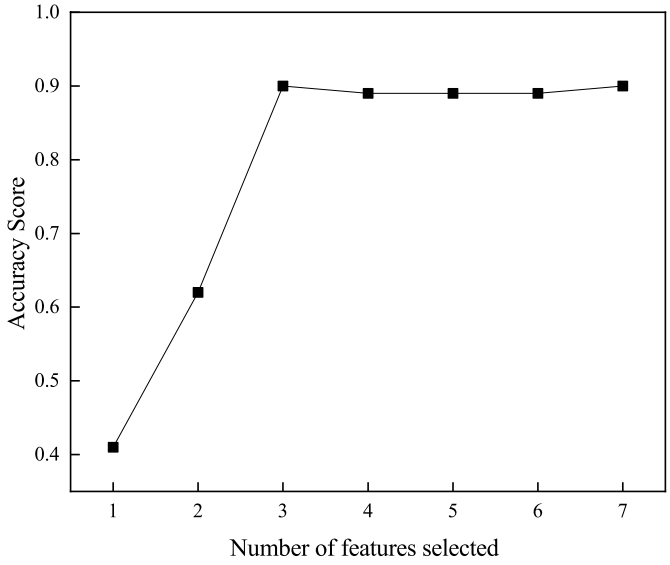


Fig. 10. Accuracy scores for different numbers of selected features.

4. Results and discussion

4.1. Calculation results of inter-well connectivity

To accurately quantify connectivity between injection and production wells, we established the G&D-Principal Feature Injection-Production Correlation Model (Equation (3)). This model integrates the static similarity coefficients and the principal features of production dynamics (water cut, monthly gas injection volume, and oil pressure). The calculated connectivity coefficients for Well Groups 008h and 010h are presented in Fig. 11. The model's regression fit yielded favorable R-squared values for each well (Fig. 12).

The G&D-Primary Feature Injection-Production Correlation Model yielded satisfactory R-squared values for each well (Fig. 12). Combined with the results from Fig. 11, it is evident that the connectivity coefficients of production wells 010-j5, 008-1, 008-17, and 008-15 are significantly higher than those of the other production wells. The calculated connectivity coefficient results are consistent with the historical liquid production data of the aforementioned four production wells. Based on the connectivity calculation results, the inter-well connectivity diagram between injection-production wells is drawn as shown in Fig. 13. It can be seen from the figure that the arrows in the figure point from the gas injection wells to the production wells, and the longer the arrow is, the greater the connectivity coefficient is and the greater the liquid production volume of the corresponding well is. The calculation results indicate that within the same well group, there are obvious differences in the connectivity between different production wells and gas injection wells. The inter-well connectivity in Well Group 008h is significantly better than that in Well Group 010h, which is related to the sedimentary facies belt where they are located (Muhetaer et al., 2021). For Well Group 008h, the effect guidance in the directions of Well 008-2 and Well 008-14 should be strengthened, while for Well Group 010h, the gas exhaust in the direction of Well 010-j5 should be appropriately controlled so that the connectivity of the well group is more reasonable.

4.2. Result analysis and verification

4.2.1. Verification by production feature comparison

To further verify the reliability of the calculation results, a comprehensive evaluation is carried out from three aspects: production characteristics, average production, and duration. The production characteristics in the pilot research area are divided into three categories, namely Category I, Category II, and Category III(Muhetaer et al., 2021). The division situation is compared with the results of this model as shown in Table 3.

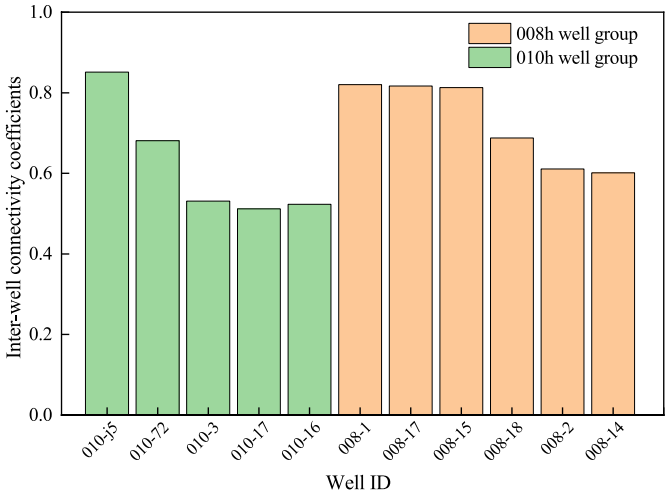


Fig. 11. Connection coefficient between production wells and gas injection wells.

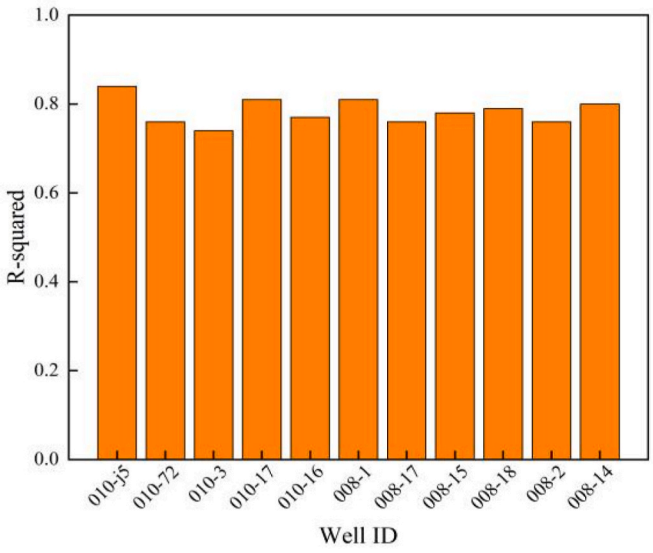


Fig. 12. The R-squared of the regression fit for each well.

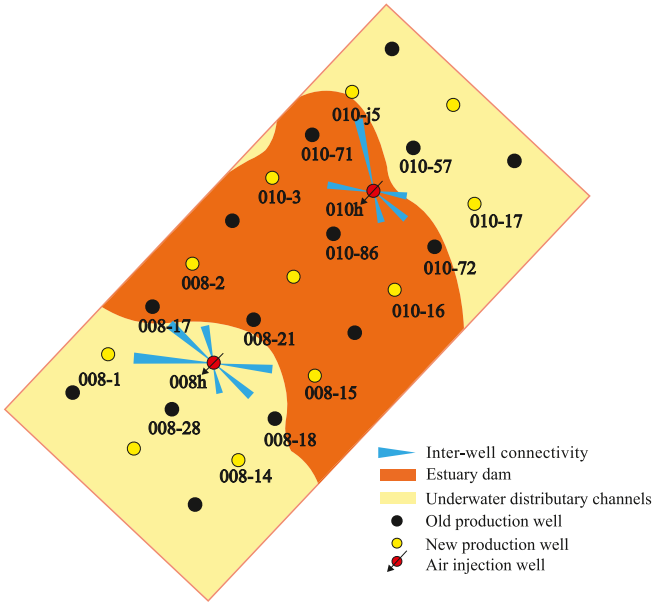


Fig. 13. Injection production connectivity diagram of some wells in pilot research area.

As shown in Table 2, wells with a connectivity coefficient above 0.8 were all classified as Category I wells, and those with a weight coefficient around 0.6 or below are all Category III wells. This indicates a positive correlation between the connectivity coefficient and production performance: higher coefficients are associated with Category I wells, while lower coefficients are associated with Category III wells. The calculation based on the G&D-Principal Feature Injection-Production Correlation Model is basically consistent with the results of well type division, indicating that the model calculation results are valid.The results of the G&D-Principal Feature Injection-Production Correlation Model show that the larger the connectivity coefficient of the inter-well connectivity in in-situ combustion is, the better the in-situ combustion production status is. Most of the connectivity calculation results of Well Group 008h are better than those of Well Group 010h, and the actual production situation also shows that the number of Category I wells is larger than that of Well Group 010h. Combined with the analysis of the

Table 3
Comparison of interwell connectivity coefficient and production well characteristics.

Well ID	Average production (t/d)	Duration (month)	Well type	Comprehensive static similarity coefficient	Inter-well connectivity coefficient	Production characteristic
010-j5	2-3	>20	I	3.11	0.851	Continuous, stable and long-term stable production
008-1					0.820	
008-15					0.813	
008-17					0.817	
008-18	1-2	10-15	II	1.85	0.688	Intermittent production with unstable output.
010-72					0.681	
008-2	<1	8-12	III	1.54	0.611	Low production, short duration and insignificant effectiveness.
008-14					0.601	
010-3					0.531	
010-16					0.523	
010-17					0.512	

geological conditions in the oilfield, Well Group 008h is located in the southwest of the overall pilot research area, where the remaining oil saturation is relatively high and the reservoir heterogeneity is not strong (Muhetaer et al., 2021). However, the reservoir heterogeneity of Well Group 010h is serious, resulting in uneven sweep of in-situ combustion and affecting the improvement of the in-situ combustion development effect.

4.2.2. Verification by tracer result comparison

Tracer testing is also a commonly used method for analyzing the inter-well connectivity situation. This method can effectively explain the affected situation corresponding to injection-production. The tracer test results of 30 days for on-site production wells are selected for verification and analysis. In Well Group 008h, Well 008-2 and Well 008-15 are selected, and in Well Group 010h, Well 010-j5 and Well 010-3 are selected. The tracer detection results of the selected wells are compared (Fig. 14).

As shown in Fig. 14(a), there is a relatively large difference in the tracer production results of the two wells. The response time and peak concentration of the tracer in Well 008-15 are both better than those in Well 008-2, so it can be judged that the connectivity of this well with the injection well is better than that of Well 008-2. Similarly, as shown in Fig. 14(b), the connectivity of Well 010-j5 is significantly better than that of Well 010-3. The injection-production connectivity coefficients of the two wells calculated in this paper also show that the connectivity status of Well 008-15 (0.813) is significantly better than that of Well 008-2 (0.611), and the connectivity of Well 010-j5 (0.851) is significantly better than that of Well 010-3 (0.531), which is consistent with the tracer results, indicating that the calculation results are relatively reasonable.

4.3. Analysis of correlation coefficients and inter-well connectivity coefficients

Through the above research, the static similarity coefficients and inter-well connectivity coefficients between injection-production wells in in-situ combustion have been calculated. The dynamic response coefficients can also be further calculated through Equation (3). In order to analyze the relationship between each indicator and the inter-well production performance, each indicator is normalized and further analyzed through the cross plot (Fig. 15).

As shown in Fig. 15, Category I wells are distributed in the upper left area of the figure, forming a relatively concentrated cluster, corresponding to the area with a relatively high static similarity coefficient (about 0.6-0.8) and a relatively low dynamic response coefficient (about 0.2 -0.4). Their distribution pattern is relatively concentrated. Category I wells show quite similar characteristics in these two indicators, that is, the degree of dynamic response is relatively low while the static similarity coefficient is relatively high.

Category II wells are distributed in the middle area of the figure and are mixed to some extent with the points of some Category III wells, being in the transition area between Category II and Category III. Both the static similarity coefficient and the dynamic response coefficient are at a medium level (about 0.4-0.5).

Category III wells are mainly distributed in the lower right area of the figure, corresponding to the area with a relatively low static similarity

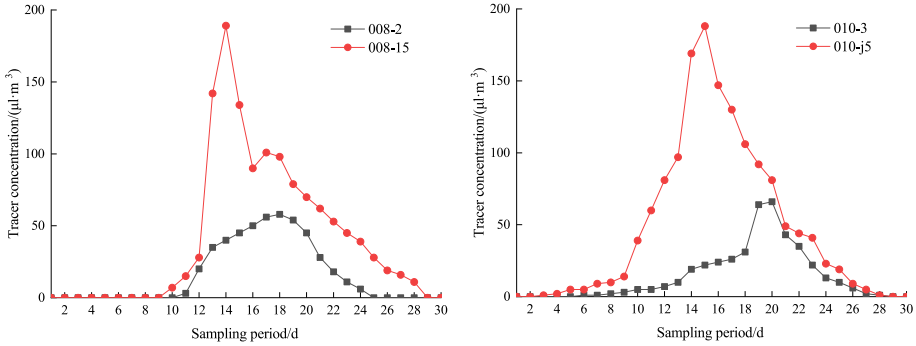


Fig. 14. Tracer test results.

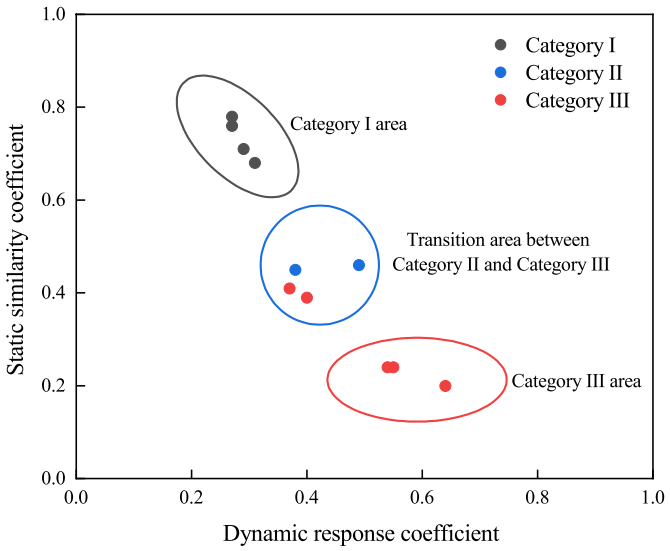


Fig. 15. Classification diagram of production wells.

coefficient (about 0.2-0.3) and a relatively high dynamic response coefficient (about 0.4-0.7).

Category III wells show relatively large differences in the above two indicators, and the static similarity coefficient is at a relatively low level. Compared with Category III wells, the production process of Category II wells is more stable, while Category III wells are prone to gas channeling and other situations, which consequently lead to a lower production volume of their production wells and even result in no production effect in some cases.

As shown in Fig. 16, as the production time increases and the amount of input data grows, the inter-well connectivity coefficients (regression coefficients of the multiple injection-production model) gradually become stable.

With the extension of the production time, the changes in the inter-well connectivity coefficients of most Category I and Category II production wells are relatively stable, and the inter-well connectivity between injection and production wells is less affected by the changes in dynamic production. This is mainly because the geological conditions between the injection and production wells are relatively good and the heterogeneity is relatively weak. In Category III wells, the changes in the connectivity coefficients are relatively drastic, mainly because the heterogeneity between the injection and production wells is relatively strong. Therefore, it is necessary to take necessary measures to adjust the state of in-situ combustion and improve the effect of in-situ combustion.

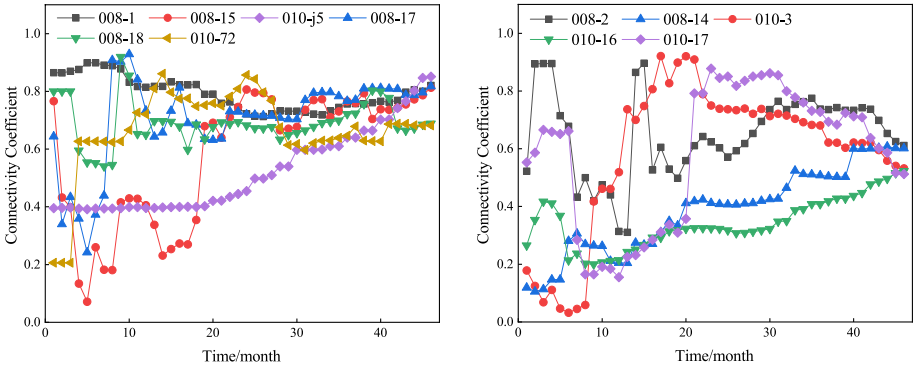
5. Conclusions

This study developed an integrated framework that combines machine learning with geological and dynamic data to enhance the analysis of inter-well connectivity in in-situ combustion, addressing the limitations of conventional methods. The principal findings and their implications are summarized as follows:

(1) A Novel Model with Practical Value. The G&D-Principal Feature Injection-Production Correlation Model effectively quantifies the impact of complex geology and production dynamics on connectivity. Its outputs show strong agreement with actual production data, providing a reliable tool for optimizing in-situ combustion strategies and improving field development planning.
(2) A Data-Driven Metric for Static Connectivity. The static similarity coefficient, derived from multiple logging parameters, offers a quantitative and efficient method to evaluate the inherent geological connectivity between wells. This approach objectively replicates the manual stratigraphic correlation process, establishing a robust static foundation for connectivity assessment.
(3) Enhanced Model Robustness through Feature Selection. By leveraging the training accuracy of multiple algorithms, key dynamic parameters (water cut, monthly gas injection volume, and oil pressure) were identified as principal features. This process reduces the risk of overfitting, and enhances model controllability and result accuracy by concentrating on the most informative parameters.
(4) Actionable Insights for Field Management.The production wells in the Hongqian In-Situ Combustion Pilot Test Area are divided into three categories of connectivity status: Category I, Category II, and Category III. Category I wells have a high static similarity coefficient and a low dynamic response coefficient. The changes in the inter-well connectivity coefficient are relatively stable, and they are less affected by the changes in dynamic production. For Category II wells, both coefficients are at a medium level. Category III wells have a low static similarity coefficient and a high dynamic response coefficient, and are prone to problems such as gas channeling. Therefore, measures should be taken to adjust the state of in-situ combustion to improve the development effect.

CRediT authorship contribution statement

Shibao Yuan: Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization. **Jia Song:** Writing – original draft, Software, Investigation, Formal analysis, Conceptualization. **Fengxiang Yang:** Writing – review & editing, Validation, Resources, Investigation.



(a) Category I and II Wells (b) Category III Wells

Fig. 16. Diagrams of the changes in inter-well connectivity of different types of wells in different time periods.

Haiyan Jiang: Writing – review & editing, Supervision, Resources, Project administration, Methodology, Conceptualization. **Zihan Ren:** Visualization, Software, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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